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May 15, 1963
SUPERSEDING
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March 30, 1955

FEDERAL SPECIFICATION

DRILL, RATCHET (REVERSIBLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hand-operated, reversible ratchet drills, generally used for drilling and tapping materials in steel construction, boiler work, and woodworking.

1.1.1 *Federal specification coverage.* Federal specifications do not include all types of the commodity indicated by the titles of the specifications but cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types.* Reversible ratchet drills shall be of the following types, as specified (see 6.1):

Type I.—American standard self-holding taper socket.
Type II.—Square taper socket.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151—Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

FSC 5110

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

American Standards Association Standard:

B5.10-1953 — Machine Tapers; Self-Holding and Steep Taper Series.

(Copies may be obtained from the American Standards Association, 70 East 45th Street, New York 17, New York.)

3. REQUIREMENTS

3.1 Illustration. The illustration herein is descriptive and not restrictive and is not intended to preclude the purchase of reversible ratchet drills otherwise conforming to this specification.

3.2 Material. The material shall conform to the requirements specified herein. Material not definitely specified shall be of a quality suitable for the intended use.

3.3 Design. The ratchet drills shall consist essentially of a handle and head, body, ratchet mechanism, feed screw, pawl, and feed pin. The ratchet drills shall be similar to figure 1 and shall conform to table I and table II, as applicable. Ratchet drills, when tested as specified in 4.3.3, shall be capable of withstanding the applicable torque loads as specified in tables I and II. Handles and tapered sockets shall show no indication of permanent set, cracks, or flaws and the ratchet mechanism and ratchet reverse control pawl engaging surfaces shall show no damage.

3.3.1 *Handle and head.* The handle and head shall be either integral, threaded, or the head shall be welded securely to the handle. The handle shall be either a straight, round steel bar, or steel tubing smoothly finished to afford a comfortable handgrip. The head shall be drop-forged steel or malleable iron and all surfaces that bear or mate with the body shall have a smooth machine finish.

3.3.2 *Body.* The body shall be of drop-forged steel with a minimum of ten teeth or notches on the outside diameter to assure ratcheting with the handle having a swing of 36 degrees or less. The inside tapered hole shall be either American Standard taper or bitstock taper to accommodate drill shanks within the capacity of the drill. The tapered hole shall be smoothly finished and shall be concentric with the bearing surfaces on the outside diameter. The outside diameter of the spindle or the end opposite the tapered hole shall be threaded to accommodate a body nut, or other means provided for securely holding the body in the head. The body shall form a part of the ratchet mechanism. The taper square hole shall be smooth and concentric with the bearing surfaces on the outside diameter.

3.3.2.1 *Sockets.* Sockets for type I ratchet drills shall have a standard self-holding taper in accordance with American Standards Association Standards B5.10. Sockets for type II ratchet drills shall be square tapered in accordance with dimensional requirements specified in table II.

3.3.3 *Ratchet mechanism.* The ratchet mechanism shall be enclosed in the head and secured in such a manner as to minimize loosening during hard usage. The ratchet mechanism shall be reversible by movement of a shifting lever, knurled button, or equally

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TABLE I.—*Type I, American Standard self-holding taper socket*

Size	Height of head	Length of feed	Length of handle (min.)	American Standard taper socket	Size of drill	Torque load (min.)	Weight (approx.)
No.	*Inches*	*Inches*	*Inches*	No.	*Inches*	*Inch-pounds*	*Pounds*
1	5	2	9	1	7/16 to 9/16	2500	1¾
2	5¼	1½	9	1	7/16 to 9/16	2500	1¾
3	6	2¼	10	2	37/64 to 29/32	3000	3½
4	6½	1¾	10	2	37/64 to 29/32	3000	3½
5	6	2¼	12	3	59/64 to 1¼	4000	5
6	6¾	2½	12	3	59/64 to 1¼	4000	5
7	8 9/16	2¾	14	3	59/64 to 1¼	5000	6½
8	6¾	2½	15	3	59/64 to 1¼	5500	6½
9	8¾	2¾	16	3	59/64 to 1¼	6000	7¼
10	9	3½	18	4	1 17/64 to 2	7000	12
11	9½	2¾	18	4	1 17/64 to 2	7000	12
12	9	3½	22	4	1 17/64 to 2	9000	12¼
13	9½	2¾	24	4	1 17/64 to 2	10,000	12¾
14	12½	3½	36	5	2 1/32 to 3	20,000	30

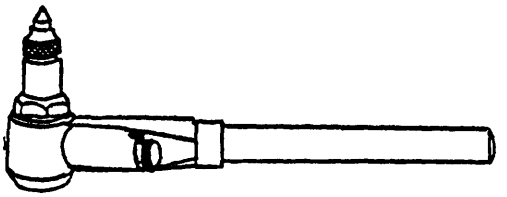


FIGURE 1.—Types I and II reversible ratchet drills, American Standard self-holding taper socket and square taper socket.

TABLE II.—*Type II, square taper socket*

Size	Height of head	Length of feed	Length of handle (min.)	Size of taper socket	Square taper	Torque load (min.)	Weight (approx.)
No.	*Inches*	*Inches*	*Inches*	*Inches*	*Number*	*Inch-pounds*	*Pounds*
1	3½	¾	9	5/16 x 5/8 x 1¼	Standard bitstock taper	2500	1¼
2	3¼	1½	9	5/16 x 5/8 x 1¼	Standard bitstock taper	2500	1½
3	4 9/16	1½	9	5/16 x 5/8 x 1¼	Standard bitstock taper	2500	1⅝
4	5	2	9	5/16 x 5/8 x 1¼	Standard bitstock taper	2500	1¾
5	3 13/16	1⅜	10	5/8 x 5/8 x 1½	1	3000	2¾
6	5 7/16	1¾	10	5/8 x 5/8 x 1½	1	3000	3½
7	4¾	1½	12	5/8 x 5/8 x 1½	1	4000	3½
8	6	2¼	12	5/8 x 5/8 x 1½	1	4000	4
9	6¾	2½	12	5/8 x 5/8 x 1½	1	4000	6
10	7 5/16	2¾	14	5/8 x 5/8 x 1½	1	5000	6¾
11	7 5/16	2¾	16	5/8 x 5/8 x 1½	1	6000	7¼
12	8⅝	2¾	18	½ x ¾ x 1¾	2	7000	12
13	9	3½	18	½ x ¾ x 1¾	2	7000	12¼
14	6	2¼	22	½ x ¾ x 1¾	2	9000	11½
15	9	3½	22	½ x ¾ x 1¾	2	9000	13
16	8⅝	2¾	24	½ x ¾ x 1¾	2	10,000	12½

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suitable means which permits ratcheting operation of the body in either direction of rotation. The shifting lever or knurled button shall be of steel of sufficient strength to assure long life under hard usage, and shall be conveniently located in the head or at the handgrip end of the handle.

3.3.4 *Feed screw.* The feed screw shall be of alloy steel. The outside diameter of the screw shall be threaded with a class 3 thread for engaging the spindle end of the body. One end of the screw shall have a knurled collar with at least four holes equally spaced at right angles to the axis to provide means for applying pressure to the drill. The knurled end of the screw shall be provided with an integral or removable hardened-steel center.

3.3.5 *Pawl.* The pawl shall be of alloy steel properly hardened. The pawl shall be firmly attached to a rod or pin suitable for reversing the action of the ratchet. A spring or other means shall be provided for applying the proper tension on the pawl when engaged in the teeth or notches of the body.

3.3.6 *Feed pin.* The feed pin shall be of steel suitable for regulating the feed screw in order to apply pressure to the drill.

3.4 Screw threads. The screw threads used with the ratchet drills shall be in conformance with the screw-thread standards of National Bureau of Standards Handbook H28.

3.5 Finish. Unless otherwise specified (see 6.1), all exposed metal surfaces of the ratchet drills shall have one of the coatings specified in 3.5.2 for protection from rust and corrosion.

3.5.1 *Surface roughness.* All external surfaces shall be free from pits, nodules, forge flash, burrs, cracks, and other deterimental defects. The forge flash shall be completely removed to blend smoothly with the adjacent surfaces. Areas that are both ground and buffed, or otherwise finished by an equivalent method and provided with a coating finish of chromium or nickel plate, shall have a surface roughness equivalent to the finish produced by 180-grit or finer abrasive. Areas not ground or buffed, or otherwise finished by an equivalent method and provided with a coating finish of chromium or nickel plate shall have a surface finish conforming to good commercial practice. Areas ground, buffed, or otherwise finished by an equivalent method and provided with a coating finish as specified under 3.5.2, other than chromium or nickel plate, shall have a surface roughness equivalent to the finish produced by 80-grit or finer abrasive. Other areas shall have a surface finish conforming to good commercial practice.

3.5.2 *Coatings.* The coatings shall be adherent, smooth, continuous and free from pits, blisters, nodules, and any other defects which would interfere with their protective value and serviceability. The minimum thickness of the coatings shall be as specified in 3.5.2.1 through 3.5.2.4 on all external visible surfaces.

3.5.2.1 *Bright chromium plate.* The coating shall be electro-deposited metals consisting of nickel followed by chromium. The minimum thickness shall be 0.0002 inch for nickel and 0.00001 inch for chromium.

3.5.2.2 *Bright nickel plate.* The coating shall be electrodeposited nickel not less than 0.0002 inch thick.

3.5.2.3 *Cadmium plate.* The coating shall be electrodeposited cadmium not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.5.2.4 *Zinc plate.* The coating shall be electrodeposited zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.5.2.5 *Hot phosphating.* The coating shall consist of chemically produced phosphate, followed with a coating of preservative.

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3.5.2.6 *Oxide coating.* The coating shall consist of a chemically produced oxide, followed with a coating of preservative.

3.6 **Identification marking.** The ratchet drills shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.7 **Workmanship.** The ratchet drills shall be clean, well made and shall be free from any defects which may affect their appearance or serviceability. All edges shall be smooth, round and shall be free from burrs or jagged edges that may cause personal injury.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order (see 6.1). The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 *Inspection of components and materials.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification.

4.2.2 *Intermediate inspection.* The external metal surfaces of the ratchet drill shall be inspected for surface roughness as specified in 4.3.1 to determine compliance with the requirements specified in 3.5.1. The inspection level shall be L8, with an AQL of 6.5 defects, expressed in defects per hundred units. Before assembly, component parts of the ratchet drills of each type shall be tested for coating thickness as specified in 4.3.1 to determine compliance with the requirements specified under 3.5.2. A lot shall consist of like components plated as a batch and under the same process and control. The sample unit shall be one component. The inspection level shall be L7 with an AQL of 4.0 defects expressed in defects per hundred units.

4.2.3 *End product inspection.*

4.2.3.1 *Visual examination.* The end product shall be visually examined for defects as outlined and classified in table III. The sample unit shall be one reversible ratchet drill. The lot size shall be expressed in units of drills for the purpose of determining the sample size in accordance with MIL-STD-105. For a lot size of 65 and under, the inspection level shall be level III with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units. For a lot size of 66 and over, the inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in defects per hundred units.

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TABLE III.—*Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Finish	Not as specified ..	X	
	Missing	X	
	Area of no film or thin film		X
	Blistered, peeled, chipped, scratched, or foreign matter imbedded		X
	Indication of rust	X	
Design	Any characteristic not in accordance with the specified requirements	X	
Construction and workmanship general	Part missing	X	
	Sharp burr or edges that may be injurious to personnel	X	
	Fractured, split, sprung, cracked, malformed, dented, bowed, broken, or otherwise impaired	X	
	Functional components that require abnormal force to operate	X	
	Any adjustable part that cannot be adjusted or is not properly adjusted to perform the functions intended	X	
	Misplaced, loose, or not in proper alignment	X	
	Not connected or joined as specified	X	
Welding and brazing (where applicable)	Missing	X	
	Completely broken or not fused	X	
	Burnt holes, slag inclusion, non-continuous where required, undercut, cracked, or partially broken	X	
	Excess weld or braze		X
Riveting (where applicable)	Missing	X	
	Loose, cracked, broken, or otherwise damaged	X	
Bolts, nuts, screws, studs and other types of threaded fasteners	Missing	X	
	Broken, stripped or fractured	X	
Assembly	Unit not properly assembled or secured or does not function as required.	X	
	Unit perceptibly out of alignment	X	
Marking identification	Missing, incomplete, not legible, not permanent or not as specified	X	

4.2.3.2 *Dimensional examination.* Examination shall be made of end product to determine compliance with dimensional requirements. Any dimension that is not within established tolerances shall be classified a defect. The inspection level shall be L8 with an AQL of 2.5 defects expressed in defects per hundred units.

4.2.3.3 *Testing of the end product.* Ratchet drills of each type shall be tested as specified in 4.3.2 to determine compliance with the requirements specified in 3.3. The sample unit shall be one reversible, ratchet drill of one type. The inspection level shall be level I with an AQL of 4.0 defects, expressed in defects per hundred units.

4.2.4 *Examination of preparation for delivery.* An examination shall be made to determine compliance with preservation and packaging, packing and marking requirements in accordance with MIL-H-15424.

4.3 Tests.

4.3.1 *Surface roughness test.* Sample specimens having a surface finished as actually

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produced by using an 80-grit or 180-grit abrasive, as applicable, shall be used for comparison testing (visual) of the component parts of the ratchet drills. The material for the specimens shall be similar to the material of the component being inspected.

4.3.2 *Coating thickness test.* Component parts of the ratchet drills shall be tested for coating thickness in accordance with the applicable test method in Section 500 of Fed. Test Method Std. No. 151.

4.3.3 *Torque load.* Ratchet drills shall be tested with the applicable torque load as specified in tables I and II. Handles shall be tested in conjunction with a tapered socket of the largest size as specified in tables I and II, for use with the handle being tested. The torque loads shall be applied with a torque producing machine, suitable lever with dead weights, or a testing machine. The load shall be applied for 1 minute and then removed. Care shall be exercised to maintain accuracy in applying the load and in measuring the effective lever arm through which the load acts. The effective lever arm, when the load is vertical and the lever is employed, is the shortest distance, in inches, between two vertical lines passing respectively through the point of load application and the center of the taper shank fitting the body opening. The specified torque load shall be the product of the effective lever arm in inches and the applied load in pounds. After application of the torque load, the ratchet drill shall be inspected to determine compliance with the requirements specified in 3.4.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of preservation, packaging, and packing for civil agency procurements.)

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.* Ratchet drills of one type and size shall be preserved and packaged in accordance with the applicable requirements of MIL-H-15424.

5.1.2 *Level C.* Cleaning, preservation, and packaging shall be in accordance with the supplier's commerical practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Levels A and B.* Ratchet drills of one type and size, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-H-15424.

5.2.2 *Level C.* Ratchet drills, packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and size required (1.2.1 and tables I and II).
(*c*) If a particular coating is required (3.5).
(*d*) Inspection record details (4.1).
(*e*) Levels of preservation, packaging, and packing (5.1 and 5.2).
(*f*) Marking, if different (5.3).

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6.2 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Drills, hand, not otherwise indexed.

Carload minimum weight 30,000 pounds.

Motor:

Tools, hand, other than power.

Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—GL
Navy—Sh
Air Force—MOA

Copies of this specification may be purchased for 5 cents each.

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